
PH102: Physics-2

Electrodynamics

Lecture # 08

Electrostatics – Special techniques:
Laplace's and Poisson's equations,
Uniqueness theorem

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D. Pamu and A. K. Sharma

ANNOUNCEMENT

Quiz – 1

Date: 06.02.2024 (Tuesday)

Time duration: 50 minutes

Venue: Respective tutorial group

Syllabus: Lecture 01 to 06 (till tutorial 4)

Mode: Subjective type (solve and write all steps)

CALCULATOR NOT ALLOWED

Previous lecture

- Obtained an expression for the work done to assemble or dismantle a charge configuration.
- In electrostatics, the energy is stored in the charge. In the context of radiation theory, it is stored in the field.
- Field inside a conductor is zero.
- Net charge resides on the surface of a conductor.
- Capacitance is a purely geometrical quantity, determined by the sizes, shapes, and separation of the two conductors.

Electrostatic boundary value problems

The electric field $\mathbf{E}$ and V can be obtained

(a) due to a charge distribution from Coulomb's law, in general,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'.$$

(b) by using Gauss's law, if the charge distribution is symmetric,

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$$

(c) by obtaining $\mathbf{E}$ from the electric potential V ,

$$\mathbf{E} = -\nabla V.$$

(d) due to a charge distribution,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$$

Drawbacks?

- Neither the charge distribution nor the potential distribution is known (true for conductors as the density is very high). Only total charge may be known.
- If V and charge distribution are known then integrals can be difficult to calculate.

Poisson's and Laplace's equation

We shall consider practical electrostatic problems where

- only electrostatic conditions (charge and potential) at some boundaries are known and it is desired to find $\mathbf{E}$ and V throughout the region,
- such problems are usually tackled using **Poisson's or Laplace's equation** or **the method of images**, and they are usually **referred to as boundary value problems**.

For a homogeneous medium (because ϵ_0 is a constant),

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{(Poisson's equation)}$$

In regions where there is no charge,

$$\nabla^2 V = 0 \quad \text{(Laplace's equation)}$$

If $\rho = 0$ **everywhere**, of course, **then $V = 0$** . There may be plenty of charge *elsewhere*, so we're confining our attention to places where there is no charge.

In Cartesian coordinates, Laplace's equation is

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

It's a fundamental formula in the sense that electrostatics can be thought of as the study of Laplace's equation. Effectively used in problems involving a set of conductors maintained at different potentials e.g., capacitors and vacuum tube diodes.

It also appears in diverse branches of physics as

- the gravitation and magnetism,
- the theory of fluid flow as a stress function,
- the theory of heat, and
- the study of soap bubbles.

In mathematics, it plays a major role in analytic function theory.

Interesting features of a second order differential equation

Let us start with $y''(x) = \frac{d^2y}{dx^2} = 0; \quad a \leq x \leq b$

Various boundary conditions lead to different results.

Boundary Condition:

No condition (General Solution)

$$y(a) = \alpha$$

$$y(a) = \alpha, \quad y(b) = \beta$$

$$y'(a) = \alpha, \quad y(b) = \beta$$

$$y'(a) = \alpha, \quad y'(b) = \beta$$

The goal is to know the

Solution

$$y = mx + c$$

$$y = m(x - a) + \alpha$$

$$y = \frac{\beta - \alpha}{b - a}x + \frac{\alpha b - \beta a}{b - a}$$

$$y = \alpha(x - b) + \beta$$

No solution

Existence of solution

Uniqueness of solution

Nature of Solution

Solution exist; Not unique

Solution exist; Not unique

Solution exist; Unique

Solution exist; Unique

Solution does not exist

for a differential equation

Laplace's equation in 1D

If V depends on only one variable, x , then the Laplace's equation is

$$\frac{d^2V}{dx^2} = 0$$

The general solution is $V(x) = mx + b$ (a straight line)

It contains two undetermined constants (m and b), as is appropriate for a second-order (ordinary) differential equation and are fixed by the boundary conditions of that problem.

Important features of the above equations are

- $V(x)$ is the average of $V(x + a)$ and $V(x - a)$, for any a :

$$V(x) = \frac{1}{2} [V(x + a) + V(x - a)]$$

Laplace's equation is a kind of averaging instruction; it tells you to assign to the point x the average of the values to the left and to the right of x .

- Laplace's equation tolerates no local maxima or minima (second derivative is zero); extreme values of V occurs at the end points (as solution is a straight line).

Laplace's equation in 2D

If V depends on two variables, say, x and y , then the Laplace's equation is

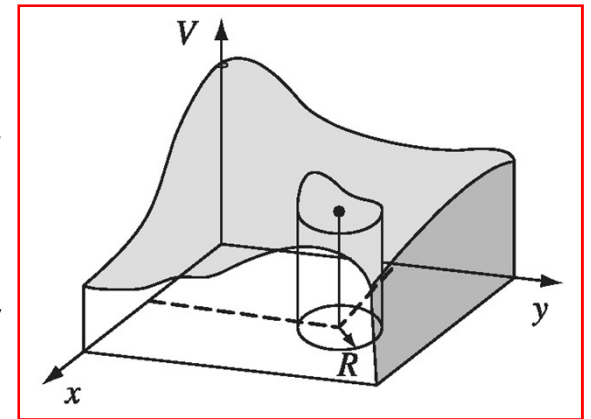
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

It is a **partial differential equation** (and not ordinary differential equation).

It is **not possible to write down the closed solution** like that for 1D equation.

So, what does the solution look like?

- Take a cardboard box, cut a wavy line all the way around, and remove the top part (see figure).
- Now glue a tightly stretched rubber membrane over the box, so that it fits like a drum head.
- Now, if you lay out coordinates (x, y) on the bottom of the box, **the height $V(x, y)$ of the sheet above the point (x, y) will satisfy Laplace's equation.**



- Harmonic functions in two dimensions have the same properties as in one dimension.
- The value of V at a point (x, y) is the average of those around the point.
- More precisely, if you draw a circle of any radius R about the point (x, y) , **the average value of V on the circle** is equal to the value at the center,

$$V(x, y) = \frac{1}{2\pi R} \oint_{\text{circle}} V dl$$

- V has no local maxima or minima; all extrema occur at the boundaries.

Laplace's equation in 3D

If V depends on three variables, x , y , and z , then the Laplace's equation is

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

- The value of V at point $\mathbf{r}$ is the average value of V over a spherical surface of radius R centered at $\mathbf{r}$,

$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} V da$$

- V can have no local maxima or minima; the extreme values of V must occur at the boundaries.

Unlike 1D and 2D cases, it is not easy to visualize or understand the solution of the 3D Laplace's equation.

Let's consider an example to prove the averaging property of the equation.

The average potential over a spherical surface

Calculate the average potential over a spherical surface of radius R due to a single point charge q located outside the sphere.

The center of the sphere is at the origin. q lies on the z -axis.

The potential at a point on the surface is

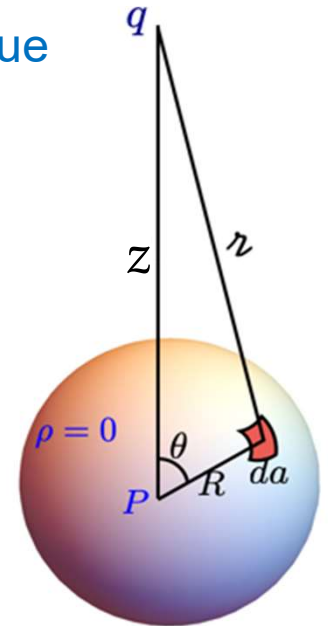
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad \text{where} \quad r^2 = z^2 + R^2 - 2zR \cos \theta$$

$$V_{\text{ave}} = \frac{1}{4\pi R^2} \frac{q}{4\pi\epsilon_0} \int [z^2 + R^2 - 2zR \cos \theta]^{-1/2} R^2 \sin \theta d\theta d\phi$$

$$= \frac{q}{4\pi\epsilon_0} \frac{1}{2zR} \sqrt{z^2 + R^2 - 2zR \cos \theta} \Big|_0^\pi$$

$$= \frac{q}{4\pi\epsilon_0} \frac{1}{2zR} [(z + R) - (z - R)] = \frac{1}{4\pi\epsilon_0} \frac{q}{z}$$

But this is precisely the potential due to q at the *center* of the sphere!!!



By the superposition principle, the same goes for any *collection* of charges outside the sphere: **their average potential over the sphere is equal to the net potential they produce at the center.**

Uniqueness Theorem

- There are several methods (analytical, graphical, numerical, experimental, etc.) of solving a given problem.
- We may wonder whether solving Laplace's equation in different ways gives different solutions.
- Therefore, before we begin to solve Laplace's equation, we should answer this question: **If a solution of Laplace's equation satisfies a given set of boundary conditions, is this the only possible solution?**
- **The answer is yes: there is only one solution. We say that the solution is unique.** Laplace's equation does not by itself determine V ; a suitable set of boundary conditions must be supplied.

Any solution of Laplace's equation which satisfies the same boundary conditions must be the only solution regardless of the method used. This is known as the *uniqueness theorem*. The theorem applies to any solution of Poisson's or Laplace's equation in a given region or closed surface.

First uniqueness theorem

The solution to Laplace's equation in some volume V is uniquely determined if V is specified on the boundary surface.

Suppose there were two solutions to Laplace's equation

$$\nabla^2 V_1 = 0 \quad \text{and} \quad \nabla^2 V_2 = 0,$$

both of which assume the specified value on the surface.

To prove that they must be equal.

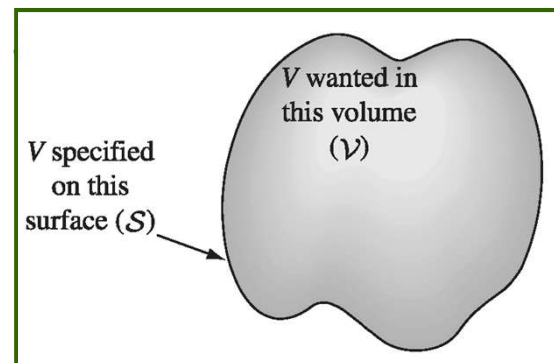
The trick is look at their *difference*: $V_3 \equiv V_1 - V_2$.

This obeys Laplace's equation (**Why?**) and therefore,

$$\nabla^2 V_3 = \nabla^2 V_1 - \nabla^2 V_2 = 0,$$

and it takes the value *zero* on all boundaries (since V_1 and V_2 are equal there).

But **Laplace's equation allows no local maxima or minima** - all extrema occur on the boundaries. So the maximum and minimum of V_3 are both zero. Therefore V_3 must be zero everywhere, and hence $V_1 = V_2$.



(Contd...)

We assumed no charge inside the region, so the potential obeyed Laplace's equation.

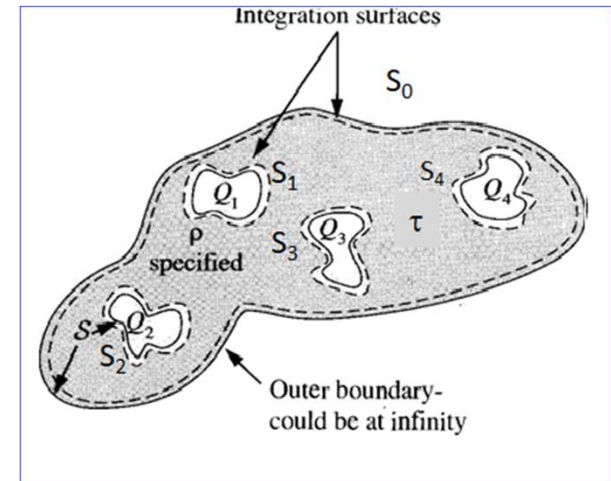
What if there is some charge inside the region?

In this case V obeys Poisson's equation.

The argument is the same, only this time

$$\nabla^2 V_1 = -\frac{1}{\epsilon_0} \rho, \quad \nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho,$$

$$\text{So, } \nabla^2 V_3 = \nabla^2 V_1 - \nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho + \frac{1}{\epsilon_0} \rho = 0.$$



Once again the *difference* ($V_3 = V_1 - V_2$) satisfies Laplace's equation and has the value zero on all boundaries, so $V_3 = 0$ and hence $V_1 = V_2$.

Corollary: The potential in a volume V is uniquely determined if

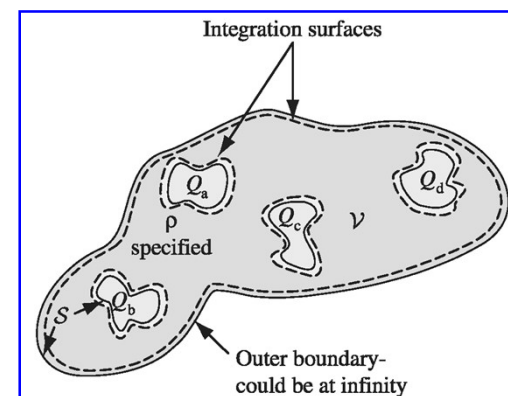
- (a) the charge density throughout the region, and
- (b) value of V on all boundaries, are specified.

Second uniqueness theorem

In a volume V surrounded by conductors and containing a specified charge density ρ , the electric field is uniquely determined if the *total charge* on each conductor is given (see figure). (The region as a whole can be bounded by another conductor, or else unbounded.)

Suppose there are *two* fields satisfying the conditions of the problem

$$\nabla \cdot \mathbf{E}_1 = \frac{1}{\epsilon_0} \rho, \quad \nabla \cdot \mathbf{E}_2 = \frac{1}{\epsilon_0} \rho.$$



Both obey Gauss's law in integral form for a Gaussian surface enclosing each conductor,

$$\oint_{i \text{ th conducting surface}} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_i, \quad \oint_{i \text{ th conducting surface}} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_i.$$

and for the outer boundary (whether this is just inside an enclosing conductor or at infinity),

$$\oint_{\text{outer boundary}} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{tot}}, \quad \oint_{\text{outer boundary}} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{tot}}.$$

The difference

$$\mathbf{E}_3 \equiv \mathbf{E}_1 - \mathbf{E}_2, \text{ which obeys } \nabla \cdot \mathbf{E}_3 = 0$$

in the region between the conductors, and

$$\oint \mathbf{E}_3 \cdot d\mathbf{a} = 0$$

over each boundary surface.

We do not know how the charge Q_i distributes itself over the i^{th} conductor, we *do* know that each conductor is an equipotential, and hence V_3 is a *constant* (not necessarily the *same* constant, it may be different on different conductors) over each conducting surface.

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$$

By taking $f = V_3$ and $\mathbf{A} = \mathbf{E}_3$

$$\nabla \cdot (V_3\mathbf{E}_3) = V_3(\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla V_3) = -(E_3)^2.$$

(first term is zero and $\mathbf{E}_3 = -\nabla V_3$.)

Therefore, from divergence theorem,

$$\int_V \nabla \cdot (V_3 \mathbf{E}_3) d\tau = \oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a} = - \int_V (E_3)^2 d\tau.$$

V_3 is a constant over each surface (if the outer boundary is infinity, $V_3 = 0$ there), so it comes outside each integral, and $\oint_S \mathbf{E}_3 \cdot d\mathbf{a} = 0$, therefore

$$\int_V (E_3)^2 d\tau = 0.$$

The only way the integral can vanish is if $\mathbf{E}_3 = 0$ everywhere. Consequently, $\mathbf{E}_1 = \mathbf{E}_2$, and the theorem is proved.

General procedures for solving Poisson's or Laplace's equation

- Solve Laplace's (if $\rho = 0$) or Poisson's (if $\rho \neq 0$) equation using either (a) direct integration when V is a function of one variable or (b) separation of variables if V is a function of more than one variable.
- The solution at this point is not unique but is expressed in terms of unknown integration constants to be determined.
- Apply the boundary conditions to determine a unique solution for V . Imposing the given boundary conditions makes the solution unique.
- Having obtained V , find $\mathbf{E}$.
- If desired, find the charge Q induced on a conductor using $Q = \int \sigma da$. If necessary, the capacitance between two conductors can be found using $C = Q/V$.

Questions?